higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# MARKING GUIDELINE

## NATIONAL CERTIFICATE

## BUILDING AND STRUCTURAL SURVEYING N4

## 12 APRIL 2019

**This marking guideline consists of 6 pages.**

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## QUESTION 1

1.1 Two equal and opposite angles✓ formed by the intersection of two straight lines✓

1.2 Curve formed by a wire, rope or chain✓ hanging freely from two points that are not in the same vertical line✓

1.3 Distance measured on an inclination (slope)✓ have to be reduced to the horizontal except in certain cases such as in some mine measuring where the plotting of results are done on the same plane on which measurements were taken✓

1.4 Temperature at which the tape was graduated, usually between 16° and 21 °C.✓ The exact temperature is supplied with the tape. The tape will be correct only at this temperature✓

1.5 Amount by which a steel tape expands or contracts uniformly throughout its whole length for a change of every 1 °C in the temperature.✓ It varies for different types of steel but it is generally between 0,000011 and 0,000013 per degree Celsius✓

(5 × 2) **[10]**

## QUESTION 2

2.1 To prepare surveying drawings such as maps, plans, cross sections or elevation drawings,✓ with profiles which have all the necessary information about natural and man-made features.✓ These are then used by the architect and the engineer to work on the design of the proposed structure.✓ The final plan is then given to the surveyor who sets out the work which might be construction projects such as roads, railways, reservoirs and building.✓ (4)

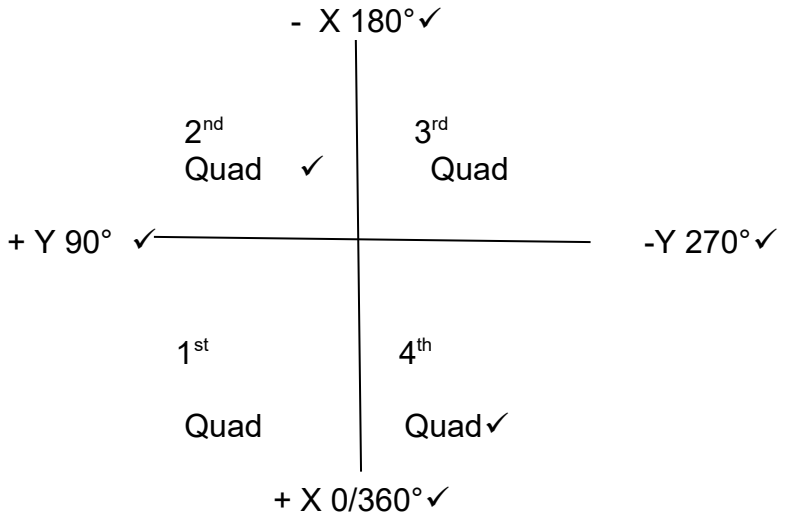
2.2
- Measuring distance and angles by handling and using survey instruments correctly
- Recording field notes accurately using standard methods
- Setting out required structures and features
- Determining elevation points
- Setting out gradients (slopes)
- Determining inaccessible points and distances (Any relevant 5 × 1) (5)

2.3
- Must be sensitive
- Magnetic axis should coincide with geometrical axis
- Ends in same horizontal and vertical planes as the pivot point
- Centre of gravity as far below pivot point as possible
- Housing box nonmagnetic (Any 5 × 1) (5)

**[14]**

**QUESTION 3**

3.1

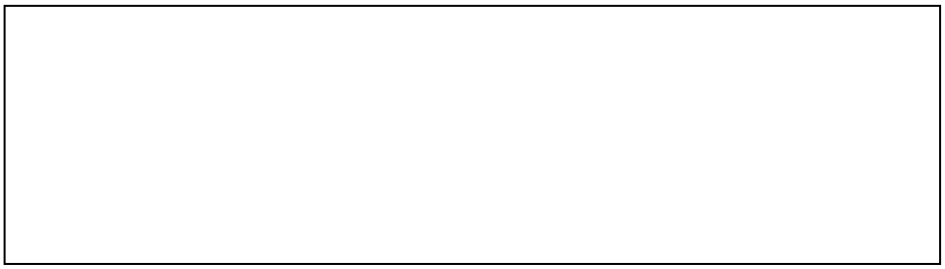



(6)

3.2
$b \times \ell = A$
$2b \times b = 60\ 000\ m^2$✓
$2b^2 = 60\ 000$
$b^2 = 30\ 000$
$b = $ ✓$30\ 000$
$b = 173,21$ m✓
$\ell = 26$✓
$\ell = 2 \times 173,21$
$\ell = 346,41$ m✓

346,41 mℓ

173,21b



(6)

3.3 Area = width/2 [first and last ordinates + 2 (sum of other ordinates)✓
= 10/2 [ (8 + 0) + 2 (7 + 9 + 6 + 5 + 3)]✓
= 5 × (8 + 60)✓
= 340 $m^2$✓ (4)

**[16]**

## QUESTION 4

4.1 grid norths;✓ reduce bearings✓ (2)

4.2 4.2.1 Quadrant bearing is the system in which the bearing line is measured eastward or westward from north or south whichever is nearer.✓ Whole-circle bearing is the bearing of a line measured in a clockwise direction from true north towards the line and varies from 0 to 360 degrees.✓

4.2.2 True bearing is the horizontal angle between true north and the survey line. It is measured from true north in a clockwise direction. Magnetic bearing is the bearing of a line measured from the magnetic north.✓

4.2.3 True north is the direction of the meridian through a point towards the geographic north pole of the earth.✓ Magnetic north is the direction in which a freely suspended compass needle points.✓

(3 × 2) (6)

4.3 4.3.1 25° is in quadrant NE and QB is the same as WCB✓
QB = N 25°$^{0}$40' E✓ (2)

4.3.2 225° is in quadrant SW✓

QB = 225° 26' – 180✓
= S 45° 26' W✓ (3)

4.3.3 140° is in the quadrant SE✓

QB = 180 – 140° 50'✓
= S 39° 50'E✓ (3)

4.3.4 340° is in the quadrant NW✓

QB = 360° – 310° 50'✓
= N 49° 10'✓ (3)

4.4 True bearing = magnetic bearing – declination✓
= 143° 40' – 8° 19'✓
= 135° 21'✓ (3)

4.5 QB is in the quadrant SE✓
WCB = 180 – 38° 45'E✓ (2)

**[24]**

## QUESTION 5

5.1 5.1.1 Rapid outward hand movements to the right side with the right hand

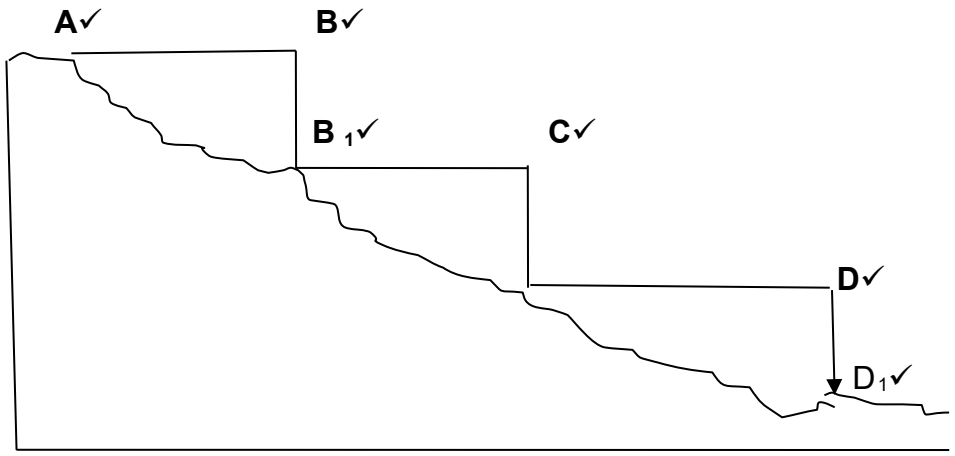
5.1.2 Right arm extended to the right, slightly bent and closed first pumped up and down

5.1.3 Both hands rose vertically above the head and kept there

5.1.4 Left arm held out horizontally to the left

(4 × 2) (8)

5.2




(6)
**[14]**

## QUESTION 6

6.1 Ct = I × e (tm – ts)
= 196,54 × 0,0000112 (11,5 – 20) ✓
= -0,0187
= 196,54 – 0,0187✓
= 196,521✓

HD = SD CosΘ
= 196,521 Cos – 7° 15'✓
= 194,950 m✓ (5)

6.2 Sc = L (I – Cos Θ)✓
= 95,45(1 – Cos 7° 40' 20")✓
= 0,854✓

CD = 95,45 – 0,854✓
= 94,496 m✓ (5)

6.3 Ch $= \frac{LH}{R}$ ✓

$= \frac{272 \times 2300}{6373\ 000}$ ✓

$= 0{,}098$ m ✓

CD $= 272 - 0{,}098$ ✓

$= 271{,}902$ m ✓ (5)

**[15]**

**QUESTION 7**

7.1 Spirit level
7.2 Eyepiece
7.3 Focus screw
7.4 Levelling screw
7.5 Trivet
7.6 Tribrach
7.7 Telescope

(7 × 1) **[7]**

**TOTAL: 100**

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